SE Lab – 1

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1. Requirements Specification

Group Travel Expense Simplification Engine — Problem Statement #37

PES University · Dept. of CSE · Lab 1: Requirements Engineering & UML Use-Case Modelling

Target Stakeholders / Actors: Group Traveler, Trip Admin (plus external Currency Exchange API)

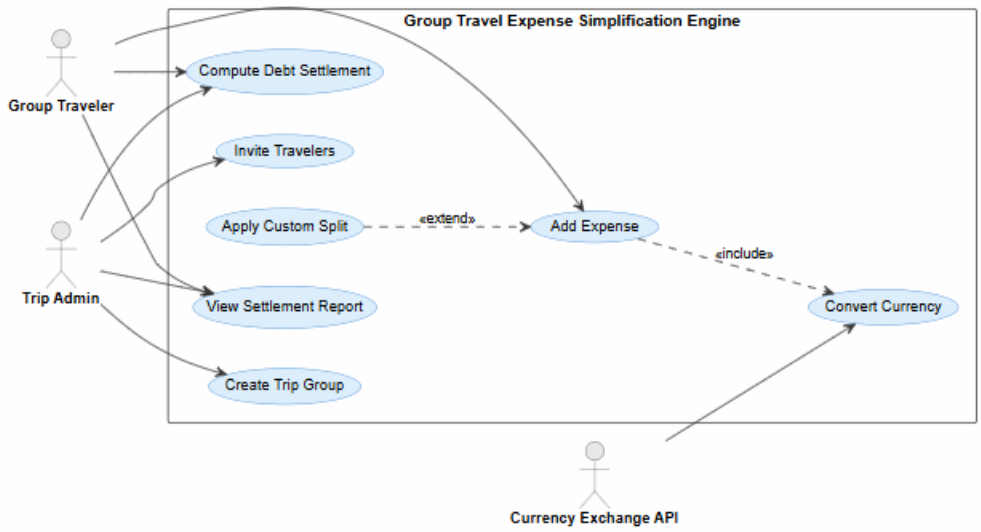
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall compute the optimal debt settlement graph using min-flow algorithms to minimize total peer-to-peer payment transactions.	High	Pass: Group balance settles to net zero with minimal payment edges. Fail: Sum of debts does not equal sum of credits.	Core value proposition of the app — reduces the number of transactions travelers must make to settle up.
FR-002	Functional	The system shall allow a Group Traveler to add an expense specifying amount, currency, payer, and participants.	High	Pass: When a traveler submits an expense with amount > 0 and at least one participant, the expense appears in the trip ledger with the correct amount, payer, and participant list. Fail: Expense is rejected or saved with missing/incorrect fields.	Core input mechanism — every downstream calculation (conversion, settlement) depends on expenses being captured accurately.
FR-003	Functional	The system shall convert every expense entered in a foreign currency to the trip's base currency using the exchange rate published on the date of entry.	High	Pass: A $50 USD expense on a EUR-based trip is stored in EUR using that day's published rate, accurate to two decimal places. Fail: Converted amount deviates from the daily published rate beyond rounding tolerance.	Multi-currency trips need a common denominator before balances can be netted and settled fairly.
FR-004	Functional	The system shall allow a Trip Admin to create a trip group, set its base currency, and invite Group Travelers by email or shareable link.	Medium	Pass: On submitting a trip name, base currency, and at least one invitee, the trip is created and each invitee receives an invitation within 60 seconds. Fail: Trip is created without a base currency, or invitees receive no invitation.	Establishes the trip container that all expenses, conversions, and settlements belong to.
FR-005	Functional	The system shall allow a Group Traveler to split an expense unevenly, by percentage or exact amount, instead of the default equal split.	Medium	Pass: When a traveler selects a custom split and enters percentages/amounts that sum to 100% / the total, the system saves the custom shares and each traveler's balance reflects them. Fail: System accepts a split that does not sum to the	Real trip expenses are rarely shared equally (e.g., one traveler orders extra food), so accurate accounting requires flexible splitting.

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				total, or silently reverts to an equal split.	
NFR-001	Nonfunctional (Performance & Security)	The settlement calculation engine must execute in under 100 ms for groups of up to 50 travelers with 500 expenses.	High	Pass: Benchmarking tests confirm target latency and security standards are met under simulated peak load. Fail: Any benchmark run exceeds 100 ms or fails a security check.	Keeps the app responsive even for large group trips, avoiding a perceived-slow experience at settle-up time.
NFR-002	Nonfunctional (Security)	The system shall encrypt all financial data (expense amounts, bank/UPI details, balances) at rest and in transit using industry-standard encryption (e.g., AES-256 at rest, TLS 1.2+ in transit).	High	Pass: A security scan confirms no financial field is stored or transmitted in plaintext, and a penetration test finds no exposed endpoints. Fail: Any financial field is found unencrypted at rest or in transit.	Financial and payment information is highly sensitive; a breach would directly harm users and erode trust in the app.

Note: *FR-001 and NFR-001 were provided in the problem statement; FR-002–FR-005 and NFR-002 were elicited from the scenario.*

2. UMLUse-Case Diagram

Actors: Group Traveler, Trip Admin, Currency Exchange API. 7 use cases (UC-01 to UC-07). Includes both required relationships: Add Expense «include» Convert Currency and Apply Custom Split «extend» Add Expense.



3. Use-Case Flow Specification

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Use Case: UC-06 — Compute Debt Settlement

Primary Actor:

Trip Admin, Group Traveler

Preconditions

- A trip group exists with a defined base currency and at least one Group Traveler.
- At least one expense has been recorded for the trip (UC-03: Add Expense).
- The requesting user (Trip Admin or Group Traveler) is authenticated and is a member of the trip.

Postconditions

Success:

A settlement graph is generated and stored, showing the minimum set of peer-to-peer payments required to bring every traveler's balance to net zero.

Failure:

No settlement graph is produced; the existing balances remain unchanged and the user is notified of the error.

Main Success Scenario

1. Trip Admin or Group Traveler selects “Settle Up” from the trip dashboard.
2. System retrieves all recorded expenses for the trip and confirms each has an amount converted to the base currency.
3. System calculates each traveler's net balance (total amount paid minus total amount owed).
4. System runs a minimum-flow settlement algorithm to compute the smallest set of peer-to-peer transactions that nets every balance to zero.
5. System displays the settlement graph, showing who owes whom and how much.
6. User reviews the settlement graph.
7. Use case ends successfully.

Alternate Flow

4a. Unconverted or Stale Expense Data

- 4a1. System detects one or more expenses without a converted base-currency amount.
- 4a2. System invokes UC-04: Convert Currency for the affected expenses.
- 4a3. If conversion succeeds, processing resumes at step 3 of the main scenario.

4a4. If conversion fails (e.g., the exchange-rate API is unavailable), the system displays an error message and the use case ends in failure.